

Amelia Eckard

Charlotte, NC | ameliaeckard.com | github.com/ameliaeckard | linkedin.com/in/ameliaeckard

Education

University of North Carolina at Charlotte <i>M.S. in Artificial Intelligence / Early Entry Program</i>	Charlotte, NC <i>Expected May 2028</i>
University of North Carolina at Charlotte <i>B.S. in Computer Science / Concentration: AI, Robotics, & Gaming</i>	Charlotte, NC <i>Expected May 2027</i>

Experience

Research Intern <i>Stevens Institute of Technology, Dr. Jina Huh-Yoo</i>	Remote <i>June 2026 – Present</i>
- Engineered <i>rq-system</i>/LEAF, an AI-assisted literature-analysis pipeline using PubMed E-utilities, CrossRef, and LLM APIs to screen, classify, and extract structured evidence from 345 papers; processed 256 included, 80 excluded, and 9 review-flagged papers. - Implemented dataset tagging, a seven-category research-question taxonomy, explicit screening/classification/extraction gates, and gold-sample validation using precision, recall, F1, accuracy, and confusion matrices.	
Undergraduate Researcher <i>UNC Charlotte, Dr. Todd Dobbs</i>	Charlotte, NC <i>August 2025 – May 2026</i>
- Developed an Apple Vision Pro indoor-navigation prototype for visually impaired users using Swift, ARKit, RealityKit, ObjectTrackingProvider, Core ML, HRTF spatial audio, and distance-based pitch cues. - Investigated object tracking, on-device recognition, and spatial-audio guidance as complementary modalities for accessible indoor navigation. Designed evaluation protocols for object identification, tracking reliability, navigation performance, and spatial-audio feedback workflows.	
Lead Instructional Assistant, ITSC 1213 <i>UNC Charlotte College of Computing and Informatics</i>	Charlotte, NC <i>August 2026 – Present</i>
Serve as lead instructional assistant for an asynchronous Python programming course, coordinating grading, rubrics, office hours, student support, and course logistics with the instructor.	

Projects

CNN Scene Classification <i>Python, PyTorch, torchvision, CUDA</i>	2026
- Trained and evaluated convolutional neural networks for 16-class scene recognition on 2,400 images. - Improved validation accuracy from 42.25% to 51.46% through image augmentation while comparing preprocessing choices, training behavior, and model performance.	
Emergency Fund Predictor <i>Python, scikit-learn, pandas, SHAP</i>	2025
- Built an 11-feature machine-learning pipeline using 12,295 Federal Reserve SHED respondents enriched with BEA indicators to study emergency-fund vulnerability. - Compared three classifiers and applied SHAP explainability to identify the strongest drivers of financial fragility and model predictions.	
Resolve – LPL Financial University Hackathon <i>AI/ML, Workflow Intelligence, FinTech</i>	2026
- Developing a proactive exception-intelligence layer that predicts likely blockers before financial-service requests visibly stall. - Designing case-history and document understanding, workflow-state comparison, anomaly and stall-risk detection, causal explanations, and next-best-action recommendations.	
Vision Lab: ASCII Reconstruction <i>JavaScript, HTML/CSS, Image Processing</i>	2026 – Present
Developed an image-to-ASCII reconstruction experiment for a web-based computer-vision lab connected to ameliaeckard.com.	

Technical Skills

Research/AI	PyTorch, scikit-learn, NumPy, pandas, OpenCV, SHAP, Core ML, LLM APIs
Methods	Precision, recall, F1, accuracy, confusion matrices, structured extraction, taxonomy design
Tools/APIs	Python, Git, CUDA, PubMed E-utilities, CrossRef, Perplexity API

Certifications & Involvement

UR2PhD Training Course, Computing Research Association	December 2025
Social and Behavioral Research, CITI Program	September 2025